

INDUSTRY PROJECT PROPOSAL

Complete all sections in full. Maximum submission: 5 pages (excluding this cover).

A. PROJECT IDENTIFICATION

A1. Project Title

VidyaConnect – Sri Lanka School Communication Platform

A2. Project Track (tick one)

☒ UCSC Track (UT)

Industry / open-source topic via
UGVLE

☐ Industry Track (IT)

PDC-registered company project

☐ Startup Track (ST)

Student-founded registered startup

A3. Project Type (tick one)

☒ Software Product

Web / mobile / desktop application

☐ Proof of Concept (PoC)

Research-based prototype / algorithm

B. GROUP & SUPERVISORY DETAILS

B1. Student Group

#	Full Name	Registration No.	Degree	Email
1	Ella Deniyage Amalsha Kavindi	2023/IS/047	☐ CS ☒ IS	edamalsha2003@gmail.com
2	Jayawarna Arachige Damsi Geli Hashara	2023/IS/031	☐ CS ☒ IS	damsijayawarna7@gmail.com
3	Hewa Wellappulige Dulmi Sehajinie	2023/IS/094	☐ CS ☒ IS	sehajiniedulmi@gmail.com
4	Fathima Zahara Preena	2023/IS/077	☐ CS ☒ IS	zaharapreena2003@gmail.com
5	Kahandha Gamage Hiruni Ravindya	2023/CS/161	☒ CS ☐ IS	ravindyahiruni7@gmail.com

i Recommended composition: at least 2 CS students and 2 IS students per group.

B2. Academic Co-Supervisor (UCSC)

Name: Ms. Rashini Lakshika

B3. Industry Mentor / Company

Name & Organisation: Mr. Ishan Liyanage

C. PROJECT OVERVIEW

C1. Problem Statement

i Describe the real-world problem or gap this project addresses. (~ 100 words)

Sri Lankan schools currently rely on fragmented and inefficient communication channels, primarily informal WhatsApp groups, printed circulars, and verbal relay through students. This approach is fundamentally ill-suited for effective school administration, leading to significant challenges. Parents frequently miss critical announcements, teachers are burdened with disproportionate communication responsibilities without delivery confirmation, and school administrators lack reliable mechanisms for distributing and tracking essential documents like consent forms and official notices. There is no single, authoritative source for school-related information, including announcements, homework deadlines, event schedules and consent requirements.

Existing global platforms often fail to address Sri Lanka's unique requirements, compatibility with low-cost Android devices, and privacy handling.

C2. Proposed Solution & Objectives

i Summarise the software solution and list 3–5 measurable project objectives.

VidyaConnect is a mobile-first, role-aware school communication platform for Sri Lankan schools, designed to replace WhatsApp groups and paper based administration in a structured and secure way.

Objectives:

1. **Develop and Deploy a Functional MVP:** Develop and deploy a mobile application (Android and iOS via React Native) and a web administration portal that support school communication, attendance management, and digital forms for at least one pilot school by the end of the project period.
2. **Implement Structured School Communication:** Enable school administrators and teachers to send targeted announcements and notifications to parents/guardians and students, with delivery tracking and notification history available within the system.
3. **Digitize Attendance and Absence Management:** Provide a digital attendance management system that allows teachers to record daily attendance, automatically notify parents/guardians of absences, and collect absence reasons with optional supporting documents.
4. **Digitize School Forms and Responses:** Replace paper-based consent and permission forms with digital forms that allow parents/guardians to submit responses online and enable administrators to monitor response status in real time.
5. **Implement Calendar Management and Integration:** Provide a unified school and class calendar that displays events, deadlines, and important activities, while allowing users to synchronize calendar events.

C3. Key Features / Modules

#	Feature / Module	Brief Description	Primary ILO(s)
1	Communication Hub	Role aware messaging replacing WhatsApp: school-wide announcements, class level messages, controlled teacher parent direct messaging, delivery confirmation, and searchable message archive.	ILO-01, ILO-02, ILO-04
2	Assignment & Task Tracking	Teachers post assignments with subject, deadline, and attachments; parents see only their child's feed; automated 24 hour deadline reminders; submission status tracking (Submitted / Late / Missing).	ILO-02, ILO-05, ILO-04
3	Smart Notification System	Sends basic notifications for assignment deadlines, upcoming events, payment confirmations, and important school announcements.	ILO-02, ILO-03, ILO-04
4	Digital Consent & Admin Forms	Admins create structured forms (Yes/No, text, date, file upload); send to parents by class/grade/school; real time response dashboard; automated reminders; Excel export of responses.	ILO-01, ILO-02, ILO-04
5	Attendance Marking & Absence Response	Enables daily attendance marking, automated absence notifications, reason submission with optional document uploads, and attendance history tracking for teachers and parents/guardians.	ILO-02, ILO-04
6	Insights Dashboard	Role specific summary dashboards: parent (assignments, messages, forms, events), teacher (attendance rates,	ILO-01, ILO-02, ILO-04

#	Feature / Module	Brief Description	Primary ILO(s)
7	Bulk Data Management	messages), admin (announcement reach, outstanding forms). Excel template bulk upload for students, parents, staff, and exam marks; bulk login creation; validation and error reporting per row; bulk export of all module data.	ILO-02, ILO-03, ILO-04

i List the main features students will individually build. Features/functions of third-party libraries cannot be claimed as contribution.

D. TECHNICAL APPROACH

D1. Technology Stack

Layer	Technology / Framework	Justification
Frontend / UI	React Native, React.js (Next.js),	Enables cross-platform mobile and web development with a modern, scalable UI.
Backend / API	Node.js + Express.js	Provides a fast, scalable, and efficient backend for application services.
Database	PostgreSQL	Ensures reliable, secure, and scalable data management.
DevOps / Deployment	GitHub Actions,Render	Supports automated deployment and simplified infrastructure management.
Other (e.g. ML, IoT)	Firebase Cloud Messaging, JWT auth, Supabase Storage	Provides push notifications, secure user authentication, and document/file storage for attendance and form submissions.

D2. System Architecture Overview

i Provide a high-level architecture diagram or a textual description of how system components interact.

VidyaConnect follows a three-tier **client-server architecture** on mobile:

Client Layer:

(a) **React Native Mobile Application (Android & iOS):**The mobile application provides access for School Administrators, Teachers, Parents/Guardians, and Students. It communicates with the backend through secure REST APIs over HTTPS and receives real-time notifications through Firebase Cloud Messaging (FCM).

(b) Next.js web portal (admin and teacher roles only). Communicates with the same REST API.

API/ Controller Layer:The Node.js and Express.js REST API exposes endpoints for authentication, user management, announcements, attendance, digital forms, and notifications. JWT-based authentication and authorization middleware secure all endpoints and enforce role-based access control. Controller-level validation verifies request structure and data integrity before forwarding requests to the service layer.

Service/Business Layer:The service layer contains the core business logic of the system. It handles application rules, permission checks, workflow execution, notification triggering, file-upload coordination, and audit logging.

Data Access Layer: The data access layer manages all interactions with the PostgreSQL database through repository or model components. It performs CRUD operations and structured queries for users, classes, attendance records, announcements, digital form responses, notifications, and audit logs.

DevOps:GitHub Actions CI/CD pipeline runs tests and deploys on merge to main. Git + GitHub for version control and code review.

Diagram Link: https://drive.google.com/file/d/1rRTvj7lSjXomNgenc1ERUP4qzC4Hw6E/view?usp=drive_link

E. PROJECT PLAN & WORKLOAD DISTRIBUTION

E1. High-Level Timeline (2 Semesters)

Phase	Milestone / Deliverable	Target Week(s)	Responsible Member(s)
S1 – 01:	Inception & Core Planning: Finalize requirements, system architecture, and initial project setup.	Weeks 1-4	All
S1 – 02:	MVP Development & Initial QA: Implement core backend APIs, mobile/web admin UI, database schema, and conduct initial testing.	Weeks 5-14	SE-focused, QA
S1 – 03:	Interim Presentation	Week 15	All
S2 – 04:	Advanced Feature Implementation: Develop all remaining advanced modules and functionalities.	Weeks 16-24	SE-focused
S2 – 05:	Conduct automated testing, security audits, performance optimization, and prepare all project documentation.	Weeks 20-29	All, QA
S2 – 06:	Final Presentation & Project Handover	Week 30	All

i Semester 1 ≈ Weeks 1–15; Semester 2 ≈ Weeks 16–30. Include Proposal, Interim, and Final milestones.

E2. Individual Contribution Plan

Student Name	Primary Responsibilities	Contribution Area	Est. Hours
1. E.D.A. Kavindi	Business Analyst, Requirements Gathering, User Story Definition, UAT Coordination, Documentation (User Manuals)	☐ SE ☒ BA ☐ Both	400 hrs
2. J.A.D.G. Hashara	Frontend Developer (Mobile App - UI/UX, Core Features), Integration Testing, Performance Optimization	☒ SE ☐ BA ☐ Both	400 hrs
3. H.W.D.Sehajinie	Backend Developer (API Endpoints, Database Integration), Security Implementation, Database Design	☒ SE ☐ BA/QA ☐ Both	400 hrs
4. F.Z. Preena	QA, Test Planning, Automated Testing (Cypress/Jest), Bug Tracking, Web Admin Portal Development	☐ SE ☒ BA/QA ☐ Both	400 hrs
5. K.G.H. Ravindya	DevOps, System Architecture, Backend Development (Complex Modules), CI/CD Implementation	☒ SE/Devops ☐ BA ☐ Both	400 hrs

i SE = Software Engineering | BA = Business Analysis. Each student must contribute ≥ 400 hours across both semesters.

F. INTENDED LEARNING OUTCOME ALIGNMENT

Tick each ILO that is directly addressed by this project and briefly explain how in the space provided.

ILO	Description	Status
ILO-01	Synthesize project requirements into structured execution plans with defined milestones and task distribution.	☒ Addressed
ILO-02	Build functional, scalable software by applying engineering principles, modern frameworks, and deployment strategies.	☒ Addressed
ILO-03	Validate software integrity through rigorous testing and maintain industry-standard technical documentation and version control.	☒ Addressed

ILO	Description	Status
ILO-04	Demonstrate professional communication and effective teamwork to achieve project objectives.	☒ Addressed
ILO-05	Resolve complex technical challenges through research and critical thinking while reflecting on professional growth.	☒ Addressed

Brief explanation of ILO coverage:

- **ILO-01:** Addressed through requirements analysis, user story creation, and development of a structured project plan with defined milestones and task allocation.
- **ILO-02:** Addressed by designing and developing VidyaConnect using modern technologies such as React Native, Next.js, Node.js, Express.js, and PostgreSQL to deliver a scalable and functional solution.
- **ILO-03:** Addressed through testing, quality assurance, documentation, and the use of GitHub for version control, code reviews, and project maintenance.
- **ILO-04:** Addressed through teamwork, collaborative development, regular meetings, stakeholder communication, and project presentations.
- **ILO-05:** Addressed by solving technical challenges such as secure data management while continuously learning and applying new technologies.

G. RISKS, CONSTRAINTS & ASSUMPTIONS

#	Risk / Constraint	Likelihood (H/M/L)	Mitigation Strategy
1	Scope Creep	Medium	Implement strict change control processes; clearly define MVP features and prioritize subsequent features; regular stakeholder meetings to manage expectations.
2	Technical Challenges (e.g. tri-lingual implementation)	Medium	Conduct thorough research and proof-of-concepts for complex features early; allocate buffer time in the schedule for R&D; leverage online communities and mentor expertise.
3	Data Privacy & Security Concerns	High	Adhere to best practices for data encryption (at rest and in transit); implement robust access control mechanisms; conduct security reviews; consult with mentor on compliance requirements.
5	Team Member Availability/Commitment	Medium	Establish clear roles and responsibilities

H. TRACK-SPECIFIC DECLARATIONS

i Complete only the sub-section relevant to your selected track (A2 above). Leave others blank.

H1. UCSC Track (UT)

UGVLE Topic Reference No.: 05	Mentor Name & Organisation: Mr. Ishan Liyanage
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H2. Industry Track (IT)

Company Name:	PDC Registration No.:	Agreement Signed? ☐ Yes ☐ No
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H3. Startup Track (ST)

Registered Company Name:	PDC Registration No.:	Founding Member(s) in Group? ☐ Yes
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I. DECLARATIONS & SIGNATURES

We confirm that the information provided is accurate, that this project is a genuine software development initiative, and that we have read and agree to the Industry Project terms and conditions.

#	Student Name	Signature	Date
1	E.D.A. Kavindi		06/02/2026
2	J.A.D.G. Hashara		06/02/2026
3	H.W.D. Sehajinie		06/02/2026
4	F.Z. Preena		06/02/2026
5	K.G.H. Ravindya		06/02/2026

Academic Co-Supervisor Name:

Ms. Rashini Lakshika

Signature & Date:



2026/06/03

Industry Mentor / Supervisor Name:

Mr. Ishan Liyanage

Signature & Date:



2026/06/03

Submit via UGVLE before the published deadline. Late submissions will not be accepted.